

Loop-R · 迭代证据检索

示例元信息

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Synthetic Research Team · 2022

This is an invented demonstration document, not a published paper.

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3 Method

Loop-R retrieves again using an intermediate answer as a reformulated query.

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6 Limitations

An incorrect intermediate answer may redirect retrieval to irrelevant passages.

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4 Evaluation

Additional retrieval rounds increase evidence coverage on two-hop questions in this controlled setting.

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Definition 1. Let U be a fixed finite evidence universe. Starting with S_0 subset U , the teaching retrieval loop updates $S_{(t+1)} = S_t \cup R_t$, where $R_t \subset U$. Previously collected evidence is never removed.

中文导读：定义第 t 轮累积证据 S_t , 更新规则为 $S_{t+1} = S_t \cup R_t$, 其中 R_t 为本轮检索结果。

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Lemma 1. Under Definition 1, every strict update increases $|S_t|$ by at least one and keeps S_t inside the fixed finite U . Proof: Definition 1 uses union with R_t subset U ; a strict update adds at least one previously absent element.

中文导读：在 Definition 1 的累积模型下, 每次严格扩展都使 $|S_t|$ 至少增加 1。

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Theorem 1. In the finite cumulative model of Lemma 1, at most $|U| - |S_0|$ strict updates can occur. Proof: Lemma 1 gives $|S_k| \geq |S_0| + k$ and $S_k \subset U$, so $k \leq |U| - |S_0|$. Repeated non-expanding retrieval is outside this bound.

中文导读：若每个继续执行的更新都至少增加一条新证据, 则成功扩展次数最多为 $|U| - |S_0|$ 。

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Corollary 1. If each retrieval terminates and the loop stops on its first non-expanding retrieval, at most $|U| - |S_0| + 1$ retrievals execute. Proof: Theorem 1 bounds strict updates by $|U| - |S_0|$; the stopping rule permits at most one final non-expanding retrieval.

中文导读：若首次无新增证据时停止, 最多执行 $|U| - |S_0| + 1$ 次检索。